Market for Loanable Funds Game

As part of this exercise, you will use the Fisher Equation to investigate the relationship between real and nominal interest rates, why expected inflation matters for lending, and the role of expectations and volatility in the clearing of borrowing and lending.

The goal of this game is to make as much profit as possible. Firms earn profit by acquiring loans from banks to complete projects. Banks earn profit by lending funds to firms and charging interest on those funds. There are 3 different types of banks and firms that have been randomly assigned to each of you. Each type of bank begins with a different, fixed supply of funds available to them to loan out, while each type of firm has a different set of projects they need to fund that generate varying amounts of profit. Firms can fund the same project multiple times. **It is in your best interests to keep this information to yourself.**

Within each group, banks and firms can negotiate with each other on a nominal interest rate and loan amount. At the end of each round, the banks collect the interest amount, while firms will receive profit from any projects they were able to fund minus the interest paid. **Interest on funds will only be paid one time, at the end of the round and no principal is to be repaid. The banks' profitability is solely a function of interest collection**. During any round, a bank or firm may choose to not give / take out loans. Any projects not funded by firms and any funds not loaned by banks earn zero profit (zero explicit penalty for holding funds). Be sure to write N/A in the sheet if that is the case.

The top 4 banks with the highest profits and the top 4 firms with the highest profits will receive +2 bonus points the first quiz. Please ensure that your name is at the top of the worksheet and can be easily read.

There are two scenarios with 5 rounds each of borrowing and lending. Funds and project funding are reset at the beginning of the second scenario. The winners will be determined by the total amount of profit across both scenarios.

More detailed instructions can be found on the back.

Scenario 1 Instructions:

- Following the available funds available to you or the funds necessary for available projects, negotiate with other people in your group on a loan amount and nominal interest rate. *Banks can only lend to one firm per round and firms can only borrow funds from one bank per round.*
- Fill in the agreed upon nominal interest rate and the loan amount in the group.
- Once all groups have finished the steps above, I will reveal the actual rate of inflation in the economy (randomized by an 8-sided dice roll).
- Using the nominal interest rate + announced inflation, calculate the real interest rate for the round.
- Determine real interest paid and resultant profit for the end of the period; record this in your worksheet.
- Repeat the above steps for four rounds.

Scenario 2 Instructions:

- The same as scenario 1, but there is a **25% chance** that one of the firms declares bankruptcy and does not repay the loan each round
- Profits earned for firms that entered bankruptcy do count towards the total